

In the Wave equation paper you showed that for the Greens function, we have that

$$\mathcal{F}G(t, \cdot)(\xi) = \frac{\sin(t|\xi|)}{|\xi|}$$

and so we see that

$$\mathcal{F}G(t, \cdot)(c\xi) = \frac{1}{c} \mathcal{F}(ct, \cdot)(\xi)$$

However in the Fractional SPDE paper the greens function is more complicated and we have that

$$\begin{aligned} \mathcal{F}Y_{\alpha, \beta, \gamma, d}(t, \cdot)(\xi) &= t^{\beta+\gamma-1} E_{\beta, \beta+\gamma}(-2^{-1} \nu t^\beta |\xi|^\alpha) \\ &= t^{\beta+\gamma-1} H_{1,2}^{1,1} \left(2^{-1} \nu t^\beta |\xi|^\alpha \middle| \begin{matrix} (0, 1) \\ (0, 1) \end{matrix} \begin{matrix} (1 - \beta - \gamma, \beta) \end{matrix} \right) \\ &= t^{\beta+\gamma-1} \frac{1}{2\pi i} \int_L \mathcal{H}_{1\frac{1}{2}}(s) \left(2^{-1} \nu t^\beta |\xi|^\alpha \right)^{-s} ds \end{aligned}$$

And so we see that

$$\mathcal{F}Y_{\alpha, \beta, \gamma, d}(t, \cdot)(c\xi) = t^{\beta+\gamma-1} \frac{1}{2\pi i} \int_L \mathcal{H}_{1\frac{1}{2}}(s) \left(2^{-1} \nu t^\beta |c\xi|^\alpha \right)^{-s} ds$$

However I am unsure how to handle the constant, c , on the inside so we can get a nice scaling property like we do in the Wave Equation paper.

Also, instead of using the Fox-H function we can just say

$$\begin{aligned} \mathcal{F}Y_{\alpha, \beta, \gamma, d}(t, \cdot)(\xi) &= t^{\beta+\gamma-1} E_{\beta, \beta+\gamma}(-2^{-1} \nu t^\beta |\xi|^\alpha) \\ &= t^{\beta+\gamma-1} \sum_{k=0}^{\infty} \frac{(-2^{-1} \nu t^\beta |\xi|^\alpha)^k}{\Gamma(\beta k + (\beta + \gamma))} \end{aligned}$$

and so

$$\mathcal{F}Y_{\alpha, \beta, \gamma, d}(t, \cdot)(c\xi) = t^{\beta+\gamma-1} \sum_{k=0}^{\infty} \frac{(-2^{-1} \nu t^\beta |c\xi|^\alpha)^k}{\Gamma(\beta k + (\beta + \gamma))}$$

but I still don't see how to handle the c .

Here is how one should proceed:

$$\begin{aligned} \mathcal{F}Y_{\alpha, \beta, \gamma, d}(t, \cdot)(c\xi) &= t^{\beta+\gamma-1} \sum_{k=0}^{\infty} \frac{(-2^{-1} \nu t^\beta |c\xi|^\alpha)^k}{\Gamma(\beta k + (\beta + \gamma))} \\ &= t^{\beta+\gamma-1} \sum_{k=0}^{\infty} \frac{(-2^{-1} \nu t^\beta c^\alpha |\xi|^\alpha)^k}{\Gamma(\beta k + (\beta + \gamma))} \\ &= t^{\beta+\gamma-1} \sum_{k=0}^{\infty} \frac{\left(-2^{-1} \nu (tc^{\frac{\alpha}{\beta}})^\beta |\xi|^\alpha \right)^k}{\Gamma(\beta k + (\beta + \gamma))} \\ &= c^{\frac{\alpha}{\beta}(1-\gamma-\beta)} (tc^{\frac{\alpha}{\beta}})^{\beta+\gamma-1} \sum_{k=0}^{\infty} \frac{\left(-2^{-1} \nu (tc^{\frac{\alpha}{\beta}})^\beta |\xi|^\alpha \right)^k}{\Gamma(\beta k + (\beta + \gamma))} \\ &= c^{\frac{\alpha}{\beta}(1-\gamma-\beta)} \mathcal{F}Y_{\alpha, \beta, \gamma, d} \left(c^{\frac{\alpha}{\beta}} t, \cdot \right) (c\xi) \end{aligned}$$

Please check if this scaling reduces to heat and wave equations correctly.